Question No : 1 of 34

The graph of an odd function is symmetrical about ----------

Answer (Please select your correct option)

- x-axis
- y-axis
- origin
- None of these

Start Time 9:41 AM

87:00

Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

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MTH631 Real Analysis II

Question No : 20 of 34

If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|dx < \infty$, then $\int_a^b f(x)dx$ is ----------

Answer (Please select your correct option)

- divergent
- conditionally convergent
- convergent
- None of these

Start Time: 9:41 AM

62:00 Time Left

20

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

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MTH631 Real Analysis II

Question No : 21 of 34

Marks: 1 **(Budgeted Time** Min)

Suppose f is a continuous and $F(x) = \int_a^x f(t)dt$ is bounded on $[a,b)$. Let g' be an absolutely integrable on $[a,b)$ and suppose $\lim_{x \to b^-} g(x) = 0$

Then----------

Answer (Please select your correct option)

- $\int_a^b f(x)g(x)dx$ converges
- $\int_a^b f(x)g(x)dx$ diverges
- $\int_a^b f(x)g(x)dx$ conditionally converges
- None of these

Start Time: 9:41 AM

37:00 Time Left

21

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 18 of 34 Marks: 1

A function f is said to be absolutely integrable on $[a,b)$ if f is locally integrable on $[a,b)$ and ----------

Answer (Please select your correct option)

- $\int_a^b |f(x)|dx < \infty$
- $\int_a^b |f(x)|dx = \infty$
- $\int_a^b |f(x)|dx \geq \infty$
- None of these

Start Time: 9:41 AM

65:00 Time Left

18

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 20 of 34 **Marks:**

If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|dx < \infty$, then $\int_a^b f(x)dx$ is _____

Answer (Please select your correct option)

- divergent
- conditionally convergent
- convergent
- None of these

Start Time: 9:41 AM

36:00 Time Left

20

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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MTH631 Real Analysis II

Question No : 24 of 34

The lower integral $\underline{\int_R} f(X)d(X)$ of f over R is the supremum of all the ———.

Answer (Please select your correct option)

- lower sums
- upper sums
- average sums
- None of these

Start Time: 9:41 AM

37:00 Time Left

24

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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MTH631 Real Analysis II

Question No : 19 of 34

An absolutely convergent integral is always ----------

Answer (Please select your correct option)

- divergent
- convergent
- conditionally convergent
- None of these

Start Time: 9:41 AM

62:00
Time Left

19

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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MTH631 Real Analysis II

Question No : 23 of 34

The upper integral $\overline{\int_R} f(X)d(X)$ of f over R is the infimum of all the----------

Answer (Please select your correct option)

- lower sums
- upper sums
- average sums
- None of these

Start Time: 9:41 AM

37:00 Time Left

23

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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MTH631 Real Analysis II

Question No : 15 of 34

If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then p is called _____

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite
- negative definite

Start Time: 9:41 AM

71:00 Time Left

15

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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Question No : 22 of 50

If f is unbounded on the nondegenerate rectangle R in $\mathbb{R}^2$, then——

Answer (Please select your correct option)

- f is not integrable on R
- f is integrable on R
- f is not differentiable on R
- None of these

Start Time: 9:41 AM

37:00 Time Left

22

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

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Question No : 12 of 34

A function of the form $L(X) = f_{x_1}(X_0)x_1 + f_{x_2}(X_0)x_2 + \ldots + f_{x_n}(X_0)x_n$ is called ———

Answer (Please select your correct option)

- quadractic function
- differential of f at X_0
- constant function
- None of these

Start Time: 9:41 AM

74:00
Time Left

12

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22

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If $h = f(x,y), x = f(r,\theta)$ and $y = f(r,\theta)$, then using the chain rule, $\frac{\partial h}{\partial \theta}$ = ...

Answer (Please select your correct option)

- $\frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta} + \frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta}$
- $\frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta}$
- $\frac{\partial f}{\partial x} + \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial \theta}$
- None of these

Start Time 9:41 AM

74:00 Time Left

13

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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MTH631 Real Analysis II

Question No : 14 of 34

If $h = f(x,y), x = r\cos\theta$ and $y = r\sin\theta$, where then using the chain rule, $\frac{\partial h}{\partial r} =$ .....

Answer (Please select your correct option)

- $\cos\theta\frac{\partial f}{\partial \theta} + \sin\theta\frac{\partial f}{\partial r}$
- $\cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y}$
- $\cos\theta\frac{\partial f}{\partial x} - \sin\theta\frac{\partial f}{\partial y}$
- None of these

Start Time: 9:41 AM

74:00 Time Left

14

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

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MTH631 Real Analysis II

Question No : 25 of 34

If f is integrable on a rectangle R, then ———

Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X) = \int_R f(X)d(X)$
- $\overline{\int_R} f(X)d(X) = \underline{\int_R} f(X)d(X) = \int_R f(X)d(X)$
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X) > \int_R f(X)d(X)$
- None of these

Start Time 9:41 AM

37:00 Time Left

25

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

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MTH631 Real Analysis II

Question No : 9 of 34

A point on the graph of a function at which the tangent line is parallel to x-axis is called the .....

Answer (Please select your correct option)

- critical point
- point of inflection
- point of relative extrema
- None of these

Start Time 9:41 AM

74:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

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MTH631 Real Analysis II

Question No : 10 of 34

The rate of change of $f(x,y)$ in the direction of a unit vector is called the ---------- of $f(x,y)$.

Answer (Please select your correct option)

- differential
- directional derivative
- mixed derivative
- None of these

Start Time: 9:41 AM

74:00
Time Left

10

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

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Question No : 11 of 34 Marks:

A function of several variables may have first order partial derivatives at a point X_0 but ---------- continuous at X_0.

Answer (Please select your correct option)

- fail to be
- must be
- always not
- None of these

Start Time: 9:41 AM

74:00
Time Left

11

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26

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MTH631 Real Analysis II

Question No : 28 of 34

A ---------- surface in $\mathbb{R}^n$ has zero content.

Answer (Please select your correct option)

- parabolic
- elliptical
- differentiable
- None of these

Start Time: 9:41 AM

37:00 Time Left

28

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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MTH631 Real Analysis II

Question No : 27 of 34

Any bounded set E with only finitely many ---------- has zero content.

Answer (Please select your correct option)

- limit points
- neighbourhood points
- interior points
- None of these

Start Time: 9:41 AM

37:00
Time Left

27

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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Question No : 30 of 34

A set of points of the form $X = X_0 + tU, t_1 \leq t \leq t_2$ is called .....

Answer (Please select your correct option)

- the neighbourhood.
- line segment.
- the closure.
- None of these

Start Time: 9:41 AM

38:00 Time Left

30

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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Question No : 3 of 50

_____ is the only region in R.

Answer (Please select your correct option)

- Set of finite points of R
- Interval
- Any finite sequence of R
- None of these

Start Time: 9:41 AM

81:00
Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

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MTH631 Real Analysis II

Question No : 4 of 34

For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ----

Answer (Please select your correct option)

- $\frac{1}{2}$
- $-\frac{1}{2}$
- 0
- 1

Start Time: 9:41 AM

80:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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MTH631 Real Analysis II

Question No : 5 of 34 Mark

If f and g are continuous functions at the point X_0 in R^*, then $\frac{f}{g}$ is continuous at X_0 provided that _____

Answer (Please select your correct option)

- $g(X_0) = 0$
- $g(X_0) \neq 0$
- $f(X_0) = g(X_0) = 0$
- None of these

Start Time: 9:41 AM

78:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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MTH631 Real Analysis II

Question No : 29 of 34

If f is integrable on disjoint sets S_1 and S_2 then f is integrable on

Answer (Please select your correct option)

- $S_1 \cup S_2$
- S_1 / S_2
- S_2 / S_1
- None of these

Start Time: 9:41 AM

38:00 Time Left

29

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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MTH631 Real Analysis II

Question No : 2 of 34

If H is an open covering of a compact subset S then, S can be covered by ----------

Answer (Please select your correct option)

- infinitely many sets from H
- finitely many sets from H
- atleast one set from H
- None of these

Start Time: 9:41 AM

83:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

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MTH631 Real Analysis II

Question No : 31 of 34

Prove that $d(S) = d(\overline{S})$ for any set S in R^1 where $d(S)$ represents the diameter of S.

Answer (Please click here to Add Answer)

Start Time: 9:41 AM

25:00 Time Left

31

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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MTH631 Real Analysis II

Question No : 32 of 34

Define a locally integrable function. Also show that the function $f(x) = \log x$ is locally integrable on $[0,1)$.

Answer (Please click here to Add Answer)

Start Time: 9:41 AM

25:00
Time Left

32

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

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MTH631 Real Analysis II

Question No : 32 of 34 **Marks:**

Define a locally integrable function. Also show that the function $f(x)=\log x$ is locally integrable on $[0,1)$.

Answer (Please click here to Add Answer)

Start Time 9:41 AM

61:00

Time Left

32

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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MTH631 Real Analysis II

Question No : 33 of 34 Marks: 5 (Budgeted Time M

Show that $(0,0)$ is a critical point of the function $p(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$ and it has positive semidefinite second differentials at $(0,0$

Answer (Please click here to Add Answer)

100%

Start Time: 9:41 AM

62:00

Time Left

33

Question Summary : (Attempted Question)

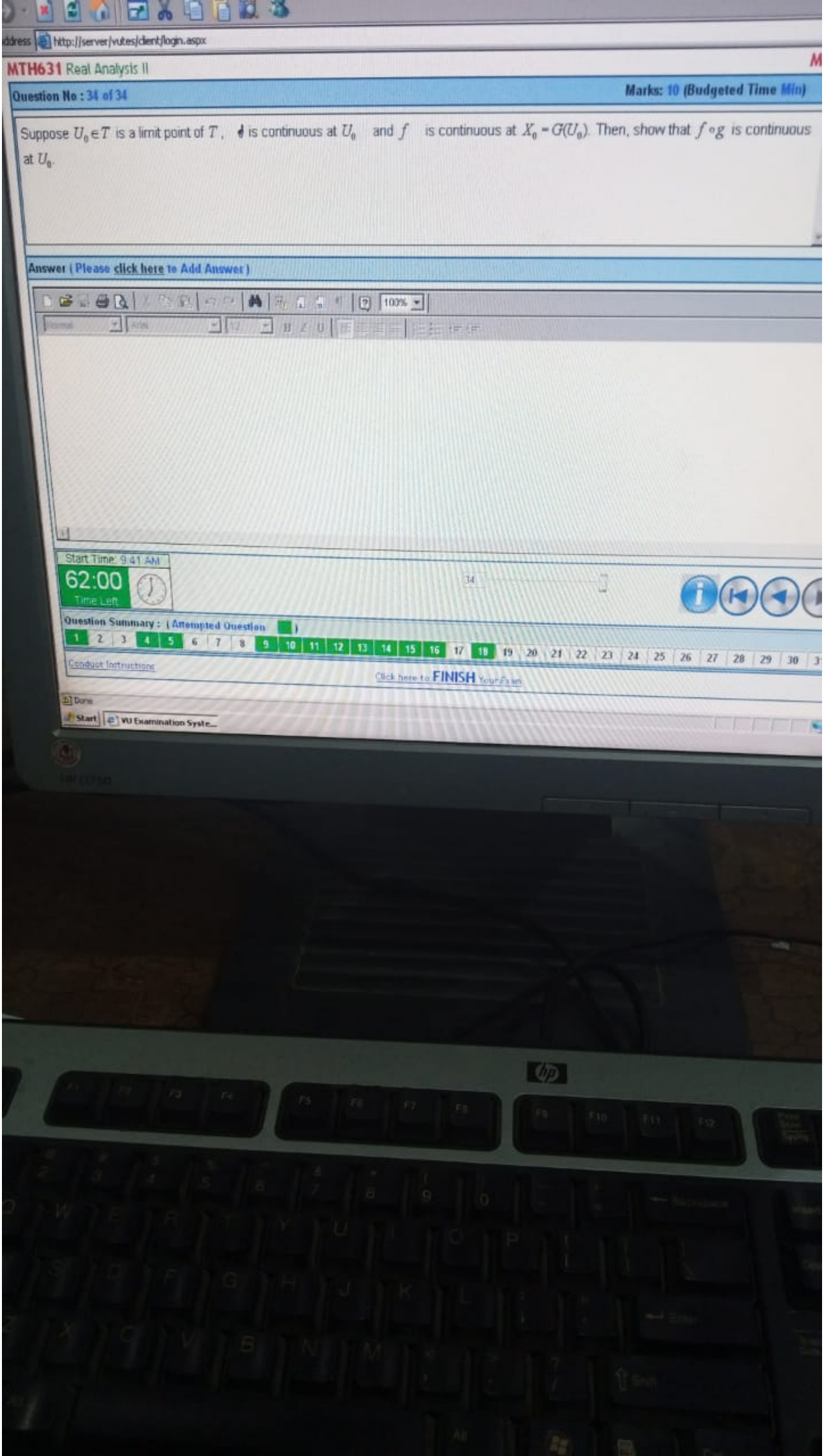
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MTH631 Real Analysis II
Question No : 34 of 34
Marks: 10 (Budgeted Time Min)
Suppose $U_0 \in T$ is a limit point of T, g is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then, show that $f \circ g$ is continuous at U_0.
Answer (Please click here to Add Answer)
Start Time: 9:41 AM
62:00
Time Left
Question Summary : (Attempted Question)
Conduct Instructions
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Done
Start
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MTH631 Real Analysis II

Question No : 29 of 34

If f is integrable on disjoint sets S_1 and S_2 then f is integrable on

Answer (Please select your correct option)

- $S_1 \cup S_2$
- S_1 / S_2
- S_2 / S_1
- None of these

Start Time 1:43 AM

74:00
Time Left

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MTH631 Real Analysis II

Question No : 28 of 34

A ---------- surface in $\mathbb{R}^n$ has zero content.

Answer (Please select your correct option)

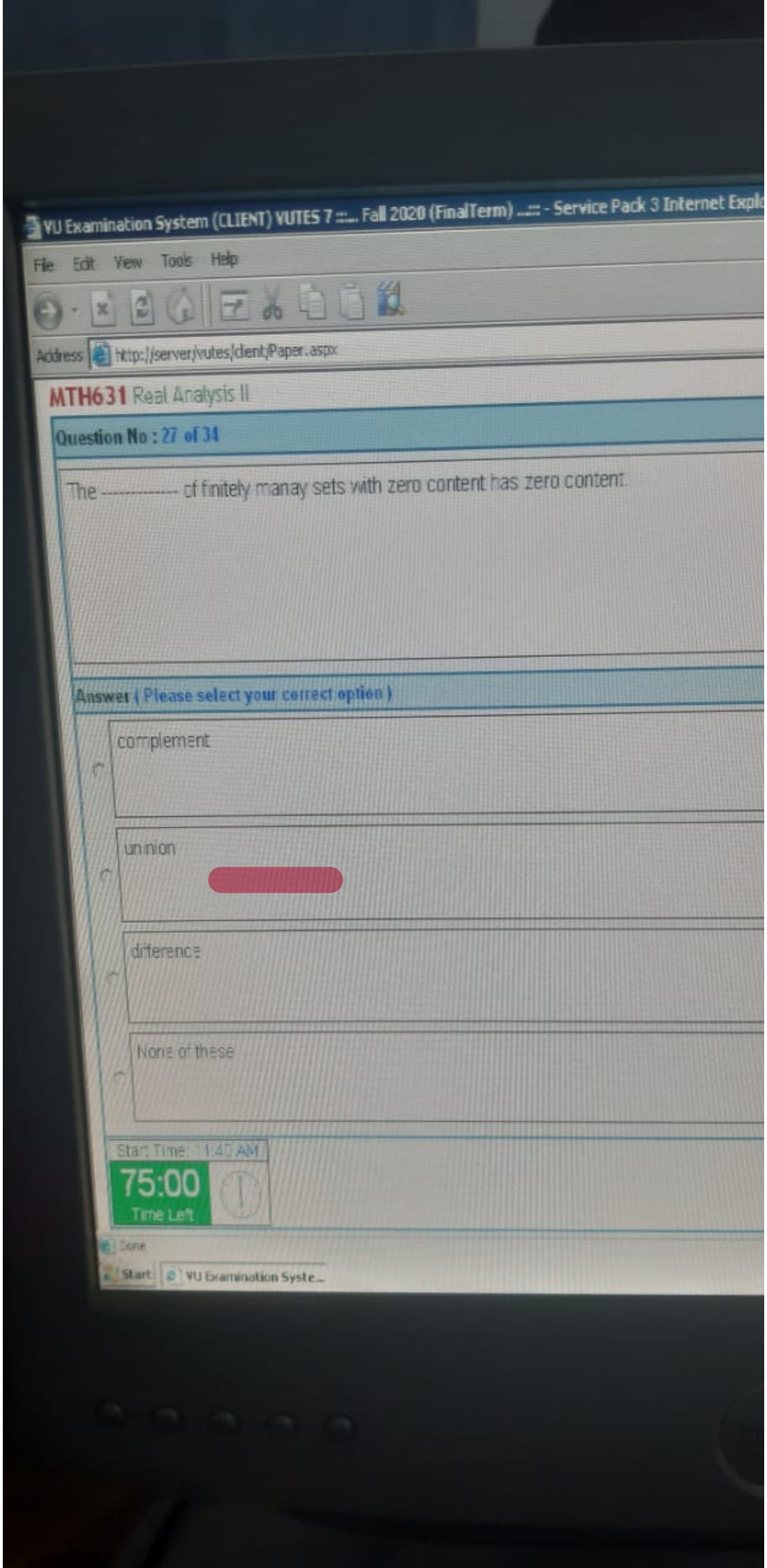
- parabolic
- eliptical
- differentiable
- None of these

Start Time: 1:40 AM

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Time Left

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MTH631 Real Analysis II
Question No : 27 of 34
The ---------- of finitely manay sets with zero content has zero content.
Answer (Please select your correct option)
complement
unnion
difference
None of these
Start Time: 1:40 AM
75:00
Time Left
Done
Start
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MTH631 Real Analysis II

Question No : 26 of 34

A bounded function f is integrable on a rectangle R if and only if ----------

Answer (Please select your correct option)

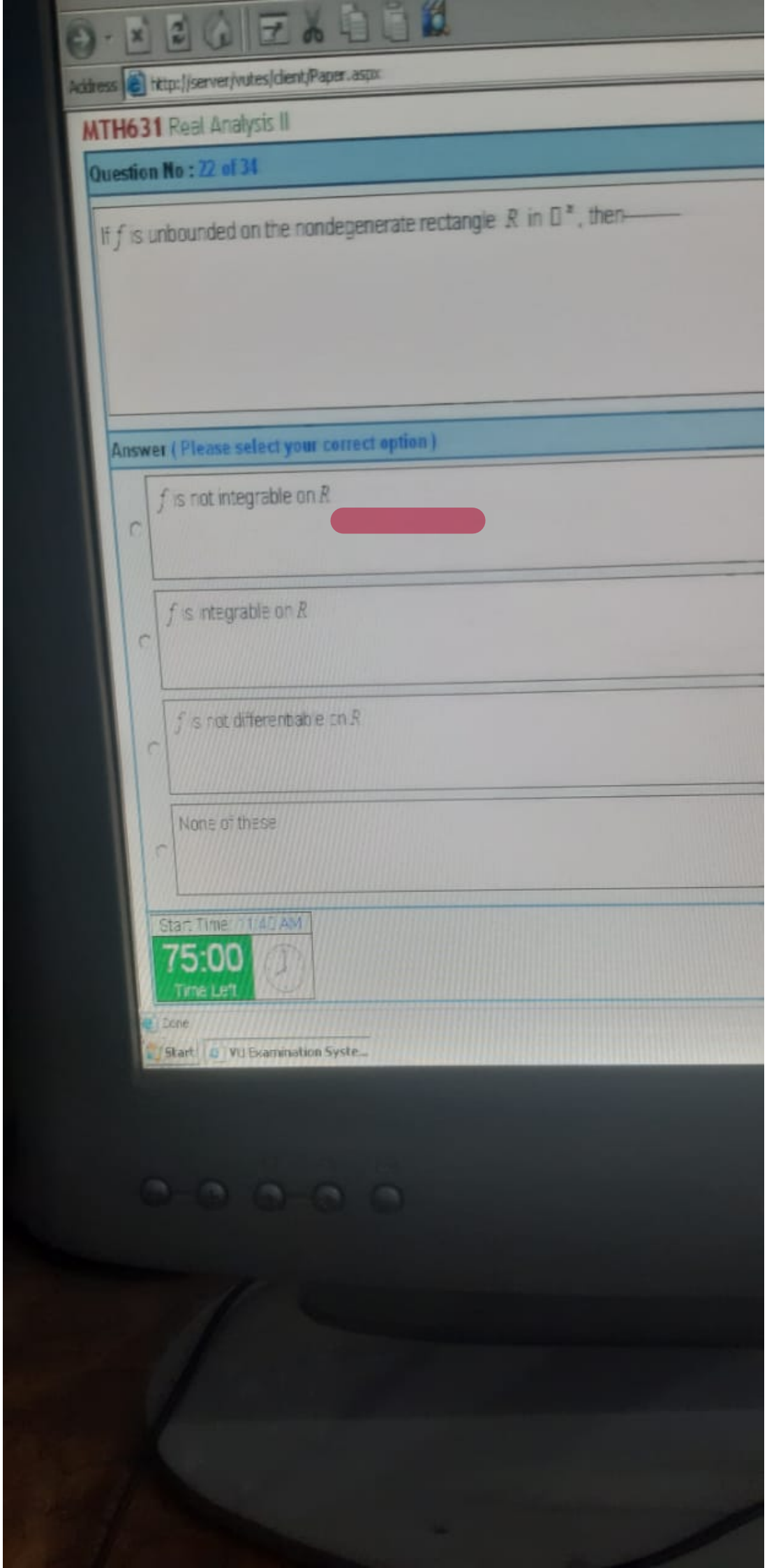
- $\overline{\int_R} f(X)d(X) = \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) < \underline{\int_R} f(X)d(X)$
- None of these

Start Time: 1:42 AM

75:00 Time Left

Done

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Address http://server/vutes/client/Paper.aspx
MTH631 Real Analysis II
Question No : 22 of 34
If f is unbounded on the nondegenerate rectangle R in $\mathbb{R}^n$, then
Answer (Please select your correct option)
f is not integrable on R
f is integrable on R
f is not differentiable on R
None of these
Start Time: 11:40 AM
75:00
Time Left
Done
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MTH631 Real Analysis II

Question No : 14 of 34

If value of a becomes zero in the Taylor series $\sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$, then the series is called ------

Answer (Please select your correct option)

- Maclaurin series
- Fourier series
- Power series
- None of these

Start Time: 11:45 AM

77:00
Time Left

14

Done

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TH631 Real Analysis II

Question No : 20 of 34

If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|dx < \infty$, then $\int_a^b f(x)dx$ is ----

Answer (Please select your correct option)

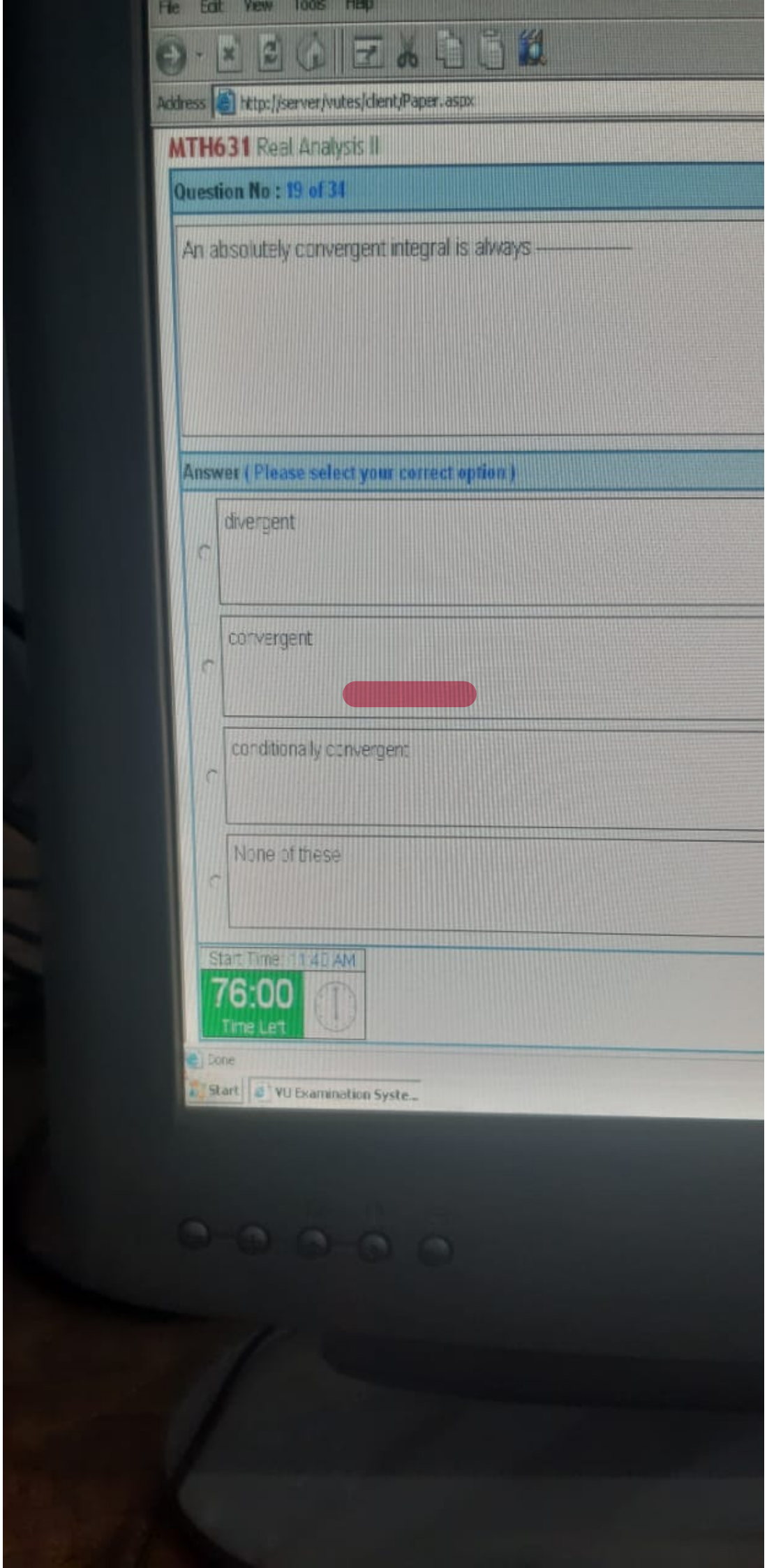
- divergent
- conditionally convergent
- convergent
- None of these

Start Time: 1:40 AM

76:00
Time Left

Done

Start | VU Examination Syste...


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MTH631 Real Analysis II
Question No : 19 of 34
An absolutely convergent integral is always ———
Answer (Please select your correct option)
divergent
convergent
conditionally convergent
None of these
Start Time: 11:40 AM
76:00
Time Left
Done
Start
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MTH631 Real Analysis II

Question No : 24 of 34

If f is bounded on a rectangle R, then ----------

Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) \leq \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) \geq \underline{\int_R} f(X)d(X)$
- None of these

Start Time: 1:40 AM

47:00 Time Left

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MTH631 Real Analysis II

Question No : 21 of 34

Suppose f is a continuous and $F(x) = \int_a^x f(t)dt$ is bounded on $[a,b)$. Let g' be an

Answer (Please select your correct option)

- $\int_a^b f(x)g(x)dx$ converges
- $\int_a^b f(x)g(x)dx$ diverges
- $\int_a^b f(x)g(x)dx$ conditionaly converges
- None of these

Start Time 1:40 AM

47:00
Time Left

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MTH631 Real Analysis II

Question No : 20 of 34

If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|dx < \infty$, then $\int_a^b f(x)dx$ is ----------

Answer (Please select your correct option)

- divergent
- conditionally convergent
- convergent
- None of these

Start Time : 1:42 AM

48:00

Time Left

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MTH631 Real Analysis II

Question No : 24 of 34

If f is bounded on a rectangle R, then ----------

Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) \leq \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) \geq \underline{\int_R} f(X)d(X)$
- None of these

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MTH631 Real Analysis II

Question No : 21 of 34

Suppose f is a continuous and $F(x) = \int_a^x f(t)dt$ is bounded on $[a,b)$. Let g be an abs

Answer (Please select your correct option)

- $\int_a^b f(x)g(x)dx$ converges
- $\int_a^b f(x)g(x)dx$ diverges
- $\int_a^b f(x)g(x)dx$ conditionally converges
- None of these

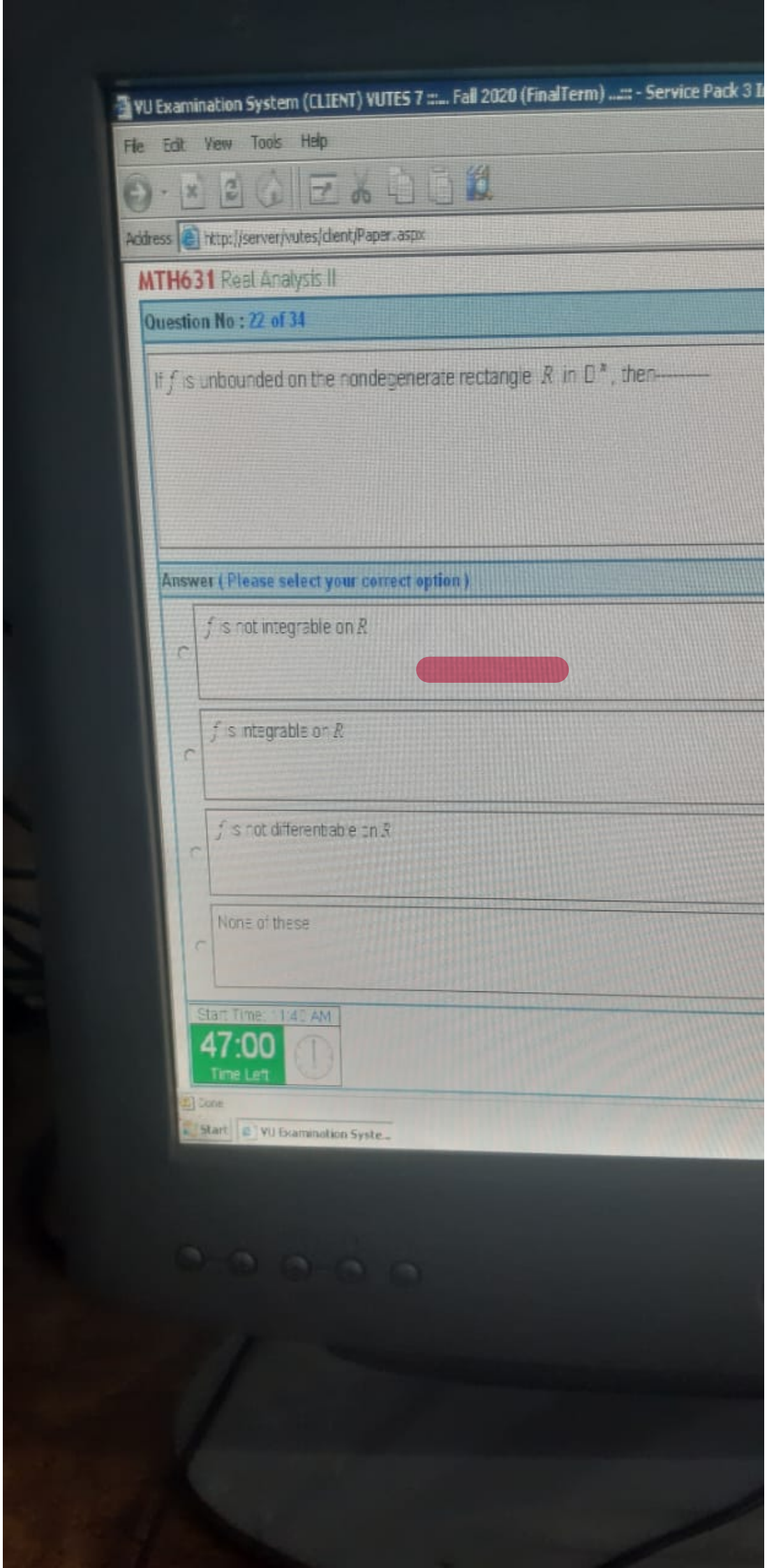
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MTH631 Real Analysis II
Question No : 22 of 34
If f is unbounded on the nondegenerate rectangle R in D*, then------
Answer (Please select your correct option)
f is not integrable on R
f is integrable on R
f is not differentiable on R
None of these
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MTH631 Real Analysis II

Question No : 30 of 34

A set is closed if and only if it contains all of its ——

Answer (Please select your correct option)

- Neighborhoods
- Limit points
- Exteriors
- None of these

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MTH631 Real Analysis II

Question No : 21 of 34

Suppose f is a continuous and $F(x) = \int_a^x f(t)dt$ is bounded on $[a,b)$. Let g' be an absolutely integrable on

Answer (Please select your correct option)

- $\int_a^b f(x)g(x)dx$ converges
- $\int_a^b f(x)g(x)dx$ diverges
- $\int_a^b f(x)g(x)dx$ conditionaly converges
- None of these

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21

Done

Start VU Examination Syste...

MTH631 Real Analysis II

Question No : 21 of 31

Suppose f is a continuous and $F(x) = \int_a^x f(t)dt$ is bounded on $[a,b)$. Let g' be an absolutely integrab

Answer (Please select your correct option)

- $\int_a^b f(x)g(x)dx$ converges
- $\int_a^b f(x)g(x)dx$ diverges
- $\int_a^b f(x)g(x)dx$ conditionaly converges
- None of these

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MTH631 Real Analysis II

Question No : 31 of 34

If X, Y and Z are in $\mathbb{R}^*$, then prove that $|X-Y| \geq ||X|-|Y||$

Answer (Please click here to Add Answer)

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MTH631 Real Analysis II

Question No : 34 of 34

Prove: If a nonempty subset of R^n is both open and closed, then $S=R^n$.

Answer (Please click here to Add Answer)

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